7

DepED
DEPARTMENT OF EDUCATION

Learning Activity Sheet in Science

Quarter 2

Lesson

8

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Quarter 2: Lesson 8 (Week 8)
S.Y. 2024-2025

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Development Team

Writers: Jhayzel P. Jocson (Leyte Normal University)
Validators: Genelita S. Garcia (Philippine Normal University)

Management Team
Philippine Normal University
Research Institute for Teacher Quality
SiMMER National Research Centre

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph.

LEARNING ACTIVITY SHEET

Learning Area:	Science	**Quarter:**	2nd Quarter
Lesson No.:	8	**Date:**	
Lesson Title/ Topic:	**Food Chain and Food Web**		
Name:		**Grade & Section:**	

I. Activity No.1: Word Hunting (10 Minutes)

II. Objective(s): unlock vocabulary words and critical thinking skills through the completion of a word puzzle.

III. Materials Needed: writing materials and worksheets

IV. Instructions:
Find the words related to the lesson in the word puzzle. Use the given clues inside the box to identify the words.

P	Z	W	O	S	T	R	O	P	H	I	C	Z	D	D
V	R	O	X	R	Y	Z	R	C	W	P	A	O	O	A
Z	P	O	T	E	O	G	Q	V	O	F	O	J	C	U
K	E	E	D	M	F	O	R	C	L	F	Y	A	D	T
O	U	L	B	U	W	Y	E	E	F	Z	R	P	E	O
M	L	W	U	S	C	L	G	A	N	N	F	R	C	T
C	H	A	I	N	M	E	Q	R	I	E	O	C	O	R
B	E	W	D	O	O	F	R	V	E	V	P	D	M	O
F	R	W	T	C	U	L	O	S	I	N	O	B	P	P
C	H	A	I	N	S	R	E	B	N	K	E	J	O	H
L	U	B	Z	P	E	G	R	V	K	H	Y	T	S	S
S	B	V	N	S	G	E	E	R	E	C	Z	B	E	I
W	F	N	N	O	H	U	B	S	K	L	V	F	R	E
M	E	T	S	Y	S	O	C	E	H	Q	W	Z	S	T
A	A	V	U	Q	O	Z	M	H	X	O	Q	A	D	S

autotrophs	carnivores	chain
chain	consumers	decomposers
ecosystem	energy	energy
flow	food	foodweb
herbivore	level	producers
trophic		

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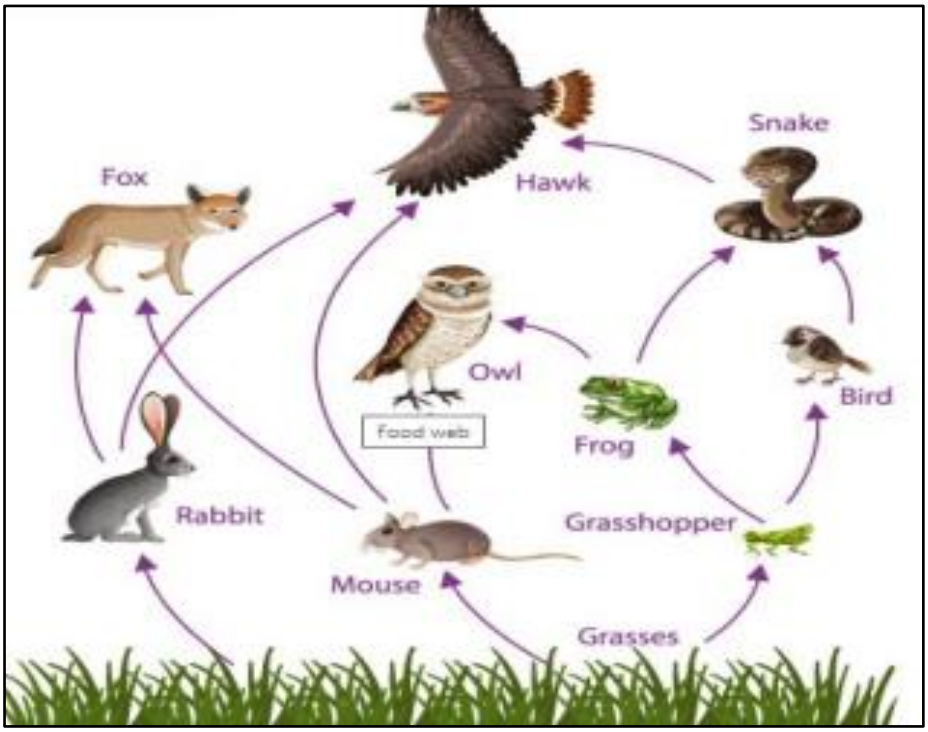
I. Activity No.2: Identifying Food Chains Within a Food Web (10 Minutes)

II. Objective(s): Identify and analyze the food chain relationships within a given food web ecosystem, discerning the flow of energy and nutrients among organisms at different trophic levels.

III. Materials Needed: writing materials, phone with internet connection and worksheets

IV. Instructions:

Examine the provided food web diagram and identify the specific food chains present within the ecosystem depicted.



Guide questions:

1. What is the producer? _____
2. What are the primary consumers? _____
3. What are the secondary consumers? _____
4. What are the top predators? _____

V. Synthesis

1. What organisms are considered producers in the food chain? What organisms are consumers in the food chain?

_____

_____

_____

2. In your everyday meal, do you prefer more meat or more vegetables? Why?

_____

_____

_____

3. What do you think is the advantage of having a meal that is mainly composed of vegetables?

_____

_____

_____

4. How much did you know about the food web before you started this activity?

_____

_____

_____

5. Did you answer the activity the way other students answered it? In what way(s) did you answer it differently or similarly?

_____

_____

_____

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I. Activity No.3: Why Taking Hurts So Bad (20 Minutes)

II. Objective(s):

a. To simulate and quantify the feeding behavior of the Philippine eagle within its forest habitat using a grid-based activity,
b. To understand the eagle's hunting efficiency and the impact of its foraging strategies on the local prey population.

III. Materials Needed:

rice grains *(about a tablespoon),* one toothpick, one sheet of bond paper, ruler, and pencil

IV. Instructions:

1. Get a clean sheet of bond paper and create a 17 x 22 grid on the paper.
2. Spread the rice grains over the grid. The paper represents the forest, the grids represent the part of the forest from which the Philippine eagle feeds, and the rice grain represents the eagle's food.
3. A Philippine eagle will catch an average of two animals per hour during daylight. Close your eyes and lower the end of the toothpick, slow down onto the grid, and imagine the eagle hunting for food.
4. Remove the grains in the square touching the toothpick. Count the grains you removed.
5. Rearrange the remaining grains on the grid and repeat steps 2 and 3 five more times and reenact one day's worth of feeding for the Philippine eagle. Count the number of total grains removed and record them in Table 1.
6. Repeat steps 2 to 4 two more times to represent three days of feeding by the Philippine eagle.

Table 1.

Day	Number of Food Caught
1	
2	
3	

7. The logging of trees causes some animals to leave their habitat and look for another place to stay. Now return all the removed rice grains to the grid. Remove 150 grains from the grid. Repeat steps 2 and 5. Record your data in Table 2.

Table 2

Day	Number of Food Caught
1	
2	
3	

V. Synthesis

1. How does the amount of food caught by a Philippine eagle relate to the changes in biotic and abiotic factors?

_____

_____

_____

_____

_____

2. How does the presence of large amounts of food allow Philippine eagles to reproduce more often?

_____

_____

_____

_____

_____

3. How would the population of other animals be affected if the amount of food remained low for a very long time?

_____

_____

_____

_____

_____

LEARNING ACTIVITY SHEET

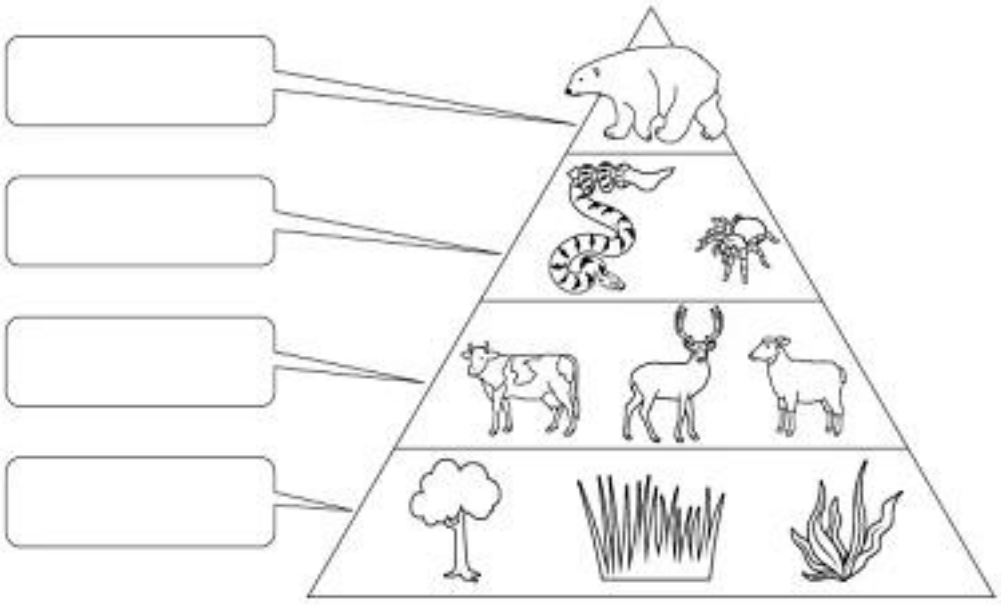
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I. Activity No.4: The Ecosystem: Feel the Energy Flow (5 Minutes)

II. Objective(s): To label and understand pyramid illustrating the flow of energy within an ecosystem

III. Materials Needed: writing materials, phone with internet connection and worksheets

IV. Instructions:
Label the different trophic levels of the energy pyramid using the word bank provided.



Producers	Tertiary consumers
Secondary consumers	Primary consumers

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I. Activity No. 4: Brain Review (5 Minutes)

II. Objective(s): To synthesize knowledge of food chain dynamics by reviewing and understanding the interactions between different trophic levels within an ecosystem.

III. Materials Needed: writing materials and worksheets

IV. Instructions:
Examine and identify the diagram below based on their role in the ecosystem.



1



2



3



4



5

LEARNING ACTIVITY SEET

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I. Activity No. 5: Ecosystem Explorer: Uncovering Food Chains in Nature (20 Minutes)

II. Objective(s): To observe and identify food chains within a local ecosystem

III. Materials Needed: field notebooks or paper, pencils or pens, magnifying glass (optional), binoculars (optional), field guidebooks or smartphone apps for identifying plants and animals

IV. Instructions:

1. Your teacher will choose a nearby natural area such as a park, forest, or field for your outdoor exploration.
2. You will be divided into small groups, each with a field notebook and writing materials.
3. Your teacher will explain the concept of food chains and how they represent the flow of energy through ecosystems.
4. Together with your group you will explore the area and observe the plants and animals present.
5. Identify different organisms and their interactions by observing who eats whom.
6. You need to record your observations in your field notebooks, including the organisms you see and the relationships you observe.
7. Identify and document at least three different food chains within the ecosystem.
8. After the exploration, your teacher will gather you by groups and each group will share their findings.
9. You will present and discuss as a class the different food chains identified and how they contribute to the overall functioning of the ecosystem.
10. Reflect on the importance of understanding food chains in ecology and how they impact the balance of nature.

Create visual representations of the food chains you observed, such as diagrams or posters.

V. Synthesis

1. How do you think these organisms are connected to each other in terms of who eats whom? Can you identify any potential predator-prey relationships?

   _____
   _____
   _____
   _____
   _____

2. How do you think energy flows through this ecosystem? Can you trace the transfer of energy from producers to consumers and eventually to decomposers?

   _____
   _____
   _____
   _____
   _____

3. How do you think energy flows through this ecosystem? Can you trace the transfer of energy from producers to consumers and eventually to decomposers?

   _____
   _____
   _____
   _____
   _____

4. How do you think disruptions to one part of a food chain might affect the rest of the ecosystem? Can you think of any examples of this occurring in nature?

   _____
   _____
   _____
   _____
   _____

5. How does understanding food chains help us appreciate the interconnectedness of life in ecosystems? What implications does this have for conservation and ecosystem management?

   _____
   _____
   _____
   _____
   _____

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I. Activity No. 6: A Comparative Journey through Aquatic and Terrestrial Worlds (10 Minutes)

II. Objective(s):

a. To create an accurate and visually appealing energy pyramid for an aquatic ecosystem, demonstrating the flow of energy from producers to consumers, with proper labeling of each trophic level.
b. To evaluate that both energy pyramids accurately represent the flow of energy within their respective ecosystems, emphasizing the roles of producers, primary consumers, secondary consumers, and other trophic levels.

III. Materials Needed:

Coloring materials, colored paper, phone with internet, writing materials and books

IV. Instructions:

1. Begin by gathering materials such as poster boards, markers, and reference materials about aquatic and terrestrial ecosystems.
2. Research and gather information about the organisms found in both aquatic and terrestrial ecosystems, focusing on their roles as producers, primary consumers, secondary consumers, and other trophic levels.
3. Use the gathered information to create two energy pyramids: one representing an aquatic ecosystem and the other representing a terrestrial ecosystem. Start by placing the primary producers at the bottom of each pyramid and arranging the subsequent trophic levels above them.
4. Label each trophic level on the energy pyramids, including the organisms found within each level and the amount of energy transferred between levels.
5. Add visual elements such as illustrations, color coding, and creative design to enhance the appeal and clarity of the energy pyramids.
6. Review the completed energy pyramids to ensure accuracy, proper labeling, organization, and creativity.
7. Practice presenting the energy pyramids, explaining the flow of energy within each ecosystem, and addressing potential questions or misconceptions that may arise.
8. Set up the visual aids in the seminar venue and be prepared to discuss the energy flow in aquatic and terrestrial ecosystems with the high school students, emphasizing the importance of wildlife conservation in maintaining ecological balance.

You are a volunteer lecturer from an international organization advocating wildlife conservation. You have been invited to conduct a free seminar for public high school students about the flow of energy in an ecosystem.

Your task is to create two energy pyramids: one in an aquatic ecosystem and the other in a terrestrial ecosystem. Your visual aid will be evaluated by a master teacher at a public high school based on creativity, proper labeling, organization, and accuracy of content

Criteria	4 points	3 points	2 points	1 point
Creativity	Visual aid demonstrates exceptional creativity, incorporating innovative design elements and engaging visuals.	Visual aid is creative and visually appealing, with some originality in design and presentation.	Visual aid shows limited creativity, with minimal variation in design and presentation.	Visual aid lacks creativity and appears dull or uninspired.
Proper Labeling	All components of the energy pyramid are clearly and accurately labeled, including producers, primary consumers, secondary consumers, etc.	Most components of the energy pyramid are labeled accurately, with only minor errors or omissions.	Some components of the energy pyramid are labeled, but there are significant errors or omissions in labeling.	Few or no components of the energy pyramid are labeled, making it difficult to understand the diagram.
Organization	Visual aid is well-organized, with a clear structure that guides the viewer's understanding of the flow of energy in the ecosystem.	Visual aid is organized, but the structure may be slightly confusing or could be improved for clarity.	Organization of the visual aid is somewhat unclear, making it challenging to follow the flow of energy in the ecosystem.	Visual aid lacks organization, making it difficult to understand the relationship between different components of the energy pyramid.
Accuracy of Content	Information presented in the visual aid is highly accurate, demonstrating a deep understanding of energy flow in both aquatic and terrestrial ecosystems.	Information presented in the visual aid is mostly accurate, with few minor inaccuracies or oversights.	Some inaccuracies or inconsistencies are present in the information presented in the visual aid, detracting from overall understanding.	Visual aid contains significant inaccuracies or misinformation that undermines its usefulness as an educational tool.